

**K. J. Somaiya College of Engineering, Mumbai-77**

**End Semester Exam**

December 2020

**Max. Marks: 30**

**Duration: 1 Hr. 15 Min. (For attempting questions) + 15 min. (uploading)**

Class: TY

Semester: V

Name of the Course: Computer Networks

Branch: Computer

Course Code:2UCC502

**Instructions:**

- (1) **All questions are compulsory**
- (2) **Draw neat diagrams**
- (3) **Assume suitable data if necessary**

| Question No. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Max. Marks   |           |                 |          |           |    |            |     |   |    |            |     |   |    |            |     |   |    |            |     |   |         |         |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------|-----------------|----------|-----------|----|------------|-----|---|----|------------|-----|---|----|------------|-----|---|----|------------|-----|---|---------|---------|
| Q.1          | List and explain design issues for OSI layers and differentiate between connection oriented and connectionless services.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 10<br>(6+4)  |           |                 |          |           |    |            |     |   |    |            |     |   |    |            |     |   |    |            |     |   |         |         |
| Q.2          | Explain working of CSMA/CA with the help of following concepts:<br>a. Inter-frame spacing<br>b. Contention window<br>c. Acknowledgements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 10           |           |                 |          |           |    |            |     |   |    |            |     |   |    |            |     |   |    |            |     |   |         |         |
|              | <p style="text-align: center;"><b>OR</b></p> <p>Compare Transport layer protocols, TCP and UDP. Explain UDP Checksum calculations with the help of an appropriate example.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 6 + 4        |           |                 |          |           |    |            |     |   |    |            |     |   |    |            |     |   |    |            |     |   |         |         |
| Q.3          | An IP router R1 implementing Classless Inter-domain routing (CIDR) receives a packet with address 131.23.151.76. Find on which port packet will be forwarded with proper explanation.<br>The router R1 routing table has the following entries:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 10           |           |                 |          |           |    |            |     |   |    |            |     |   |    |            |     |   |    |            |     |   |         |         |
|              | <table border="1"><thead><tr><th>Mask</th><th>Network Address</th><th>Next Hop</th><th>Interface</th></tr></thead><tbody><tr><td>12</td><td>131.16.0.0</td><td>---</td><td>3</td></tr><tr><td>14</td><td>131.28.0.0</td><td>---</td><td>5</td></tr><tr><td>16</td><td>131.19.0.0</td><td>---</td><td>2</td></tr><tr><td>15</td><td>131.22.0.0</td><td>---</td><td>1</td></tr><tr><td>Default</td><td>0.0.0.0</td><td>131.22.10.10</td><td>1</td></tr></tbody></table> <p>Also, Draw the neat diagram of the network with proper labelling.</p> <p style="text-align: center;"><b>OR</b></p> <p>Write Short Note on following (Any 2):</p> <ul style="list-style-type: none"><li>a. DNS</li><li>b. SMTP</li><li>c. DHCP</li></ul> |              | Mask      | Network Address | Next Hop | Interface | 12 | 131.16.0.0 | --- | 3 | 14 | 131.28.0.0 | --- | 5 | 16 | 131.19.0.0 | --- | 2 | 15 | 131.22.0.0 | --- | 1 | Default | 0.0.0.0 |
| Mask         | Network Address                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Next Hop     | Interface |                 |          |           |    |            |     |   |    |            |     |   |    |            |     |   |    |            |     |   |         |         |
| 12           | 131.16.0.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | ---          | 3         |                 |          |           |    |            |     |   |    |            |     |   |    |            |     |   |    |            |     |   |         |         |
| 14           | 131.28.0.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | ---          | 5         |                 |          |           |    |            |     |   |    |            |     |   |    |            |     |   |    |            |     |   |         |         |
| 16           | 131.19.0.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | ---          | 2         |                 |          |           |    |            |     |   |    |            |     |   |    |            |     |   |    |            |     |   |         |         |
| 15           | 131.22.0.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | ---          | 1         |                 |          |           |    |            |     |   |    |            |     |   |    |            |     |   |    |            |     |   |         |         |
| Default      | 0.0.0.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 131.22.10.10 | 1         |                 |          |           |    |            |     |   |    |            |     |   |    |            |     |   |    |            |     |   |         |         |
|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 5+5          |           |                 |          |           |    |            |     |   |    |            |     |   |    |            |     |   |    |            |     |   |         |         |